



/ABC OF PROGRAMMING

Where cool stuff starts.





/FROM STEPS TO CODE



/KEYWORDS

The words that computers understand

await	break	case	catch	class
const	continue	debugger	default	delete
do	else	enum	export	extends
false	finally	for	function	if
implements	import	in	instanceof	interface
let	new	null	package	private
protected	public	return	super	switch
static	this	throw	try	true
typeof	var	void	while	with
yield				



/FROM STEPS TO CODE



/SYNTAX

The set of rules that define a correctly structured JavaScript program

```
var a = 5;  
console.log(a); // 5  
console.log(A); // throws a ReferenceError: A is not defined
```

```
for (let i = 0; i < 10; i++) console.log(i);  
console.log(i); // throws a ReferenceError: i is not defined
```



/WRITING SCRIPTS

How do I write code?



/HOW HTML, CSS, & JS FIT TOGETHER?



/HTML

`.html files`

CONTENT LAYER

The HTML gives the page structure and adds semantics



/CSS

`.css files`

PRESENTATION LAYER

Enhances the HTML page with rules that state how the HTML content is presented (borders, colors, dimensions, etc.)



/JAVASCRIPT

`.js files`

BEHAVIOR LAYER

We can change the page behavior and adding interactivity. We'll keep JavaScript as possible in separated files.



The **JavaScript** is added last and enhances the usability of the page or the experience of interacting with the site.



/CREATING A BASIC JAVASCRIPT

INTERNALLY

```
<script>  
    // Your javascript code goes here  
</script>
```



/CREATING A BASIC JAVASCRIPT

EXTERNALLY

```
<script src="directory/filename.js"></script>
```




**/JAVASCRIPT RUNS WHERE
IT IS FOUND IN THE HTML**

\OBJECTS

How do computers fit in with the world around them?

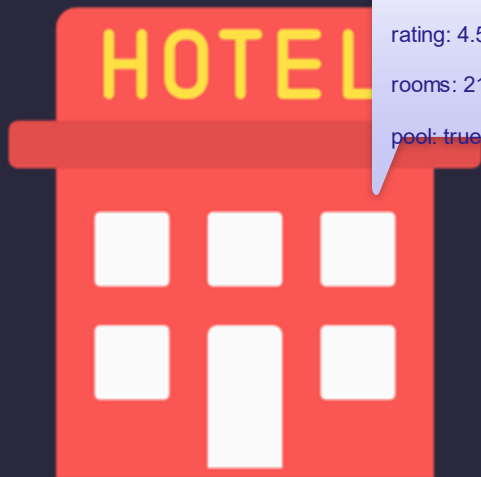




Computers create **models** of the world using
data



/OBJECT AND PROPERTIES



OBJECT TYPE:

HOTEL

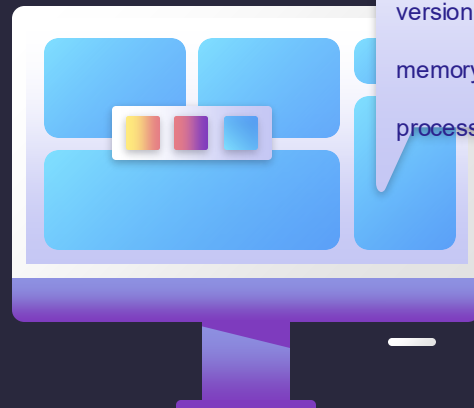
PROPERTIES

name: Quay

rating: 4.5

rooms: 21

pool: true



OBJECT TYPE:

COMPUTER

PROPERTIES

os: MacOS Mojave

version: 10.14.6

memory: 8 GB

processor: Intel Core i5



/OBJECTS



In computer programming, each physical thing in the world can be represented as an **object**.

Each object can have its own:

- ✓ Properties
- ✓ Events
- ✓ Methods

/PROPERTIES

- Properties provide details about instances of objects.
- Each property has a name and a value.
- Example:

os: MacOS Mojave

Property value

Property name

OBJECT TYPE:
COMPUTER

PROPERTIES

os: MacOS Mojave

version: 10.14.6

memory: 8 GB

processor: Intel Core i5



/OBJECTS & PROPERTIES

```
const computer = {  
  os: "MacOS Mojave",  
  version: "10.14.6",  
  memory: "8GB",  
  processor: "Intel Core i5"  
};
```

```
alert(computer.os)
```

/METHODS

Methods are **actions** that can be performed on objects

- Tell you something about that object (using information stored in its properties)

```
computer.printVersion()
```

- Change the value of one or more of the object's properties.

```
hotel.setRating(5.0)
```

```
hotel.setMemory(32GB)
```

OBJECT TYPE: **COMPUTER**

PROPERTIES

os: MacOS Mojave

version: 10.14.6

memory: 8 GB

processor: Intel Core i5

OBJECT TYPE:
HOTEL

PROPERTIES

name: Quay

rating: 4.5

rooms: 21

pool: true

/DOCUMENT OBJECT MODEL (DOM)

The current web page loaded into each window is modelled using a **document** object

- The document object is the root node of the HTML document.
- The document object is a property of the window object.
- The document object is accessed with:

`window.document` or

`document`

See more https://www.w3schools.com/jsref/dom_obj_document.asp

/HOW TO USE OBJECTS & METHODS

The diagram illustrates the components of the JavaScript code snippet `document.write('Good afternoon');`. The code is displayed in a monospace font on a dark background. The word `document` is highlighted in red, and the word `write` is highlighted in yellow. Dotted lines connect labels to specific parts of the code: `document` is labeled as the **OBJECT**, `.` is the **MEMBER OPERATOR**, `write` is the **METHOD**, and the text `('Good afternoon')` is the **PARAMETERS**.

```
document.write('Good afternoon');
```

OBJECT

MEMBER OPERATOR

PARAMETERS

METHOD

/DOM SEARCH BY ID

Using the document object we can access and change HTML elements by ID.

```
document.getElementById(id)
```

EXAMPLE:

HTML

```
<p><span id="key">Web browsers</span> use objects </p>
```

SCRIPT

```
document.getElementById("key").innerHTML = "Javascript"
```

SEARCH

UPDATE

/DOM **SEARCH** BY TAG NAME

Using the document object we can access and change HTML elements by HTML tag.

```
document.getElementsByTagName(tag)[index]
```

EXAMPLE:

HTML

```
<p><span>Web browsers</span> use objects </p>
```

SCRIPT

```
document.getElementsByTagName("span")[0].style.color = "yellow"
```

/DOM **SEARCH** BY CLASS NAME

Using the document object we can access and change HTML elements by class name.

```
document.getElementsByClassName(className)[index]
```

EXAMPLE:

HTML

```
<p><span class="key">Web browsers</span> use objects </p>
```

SCRIPT

```
var span = document.getElementsByClassName("key")[0]  
span.style.fontSize = "20px"
```

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\JS BASICS

Syntax and Grammar

{.js}

JavaScript



A **script** is a series of instructions that a computer can follow to achieve a goal.

/STATEMENTS

Each individual instruction (step) is known as a **statement**.

```
var today = new Date()
```



ASSIGNMENT: When you see the equals symbol (=), it means the value on the right is assigned to the variable in the left.



JavaScript statement

/COMMENTS

You should write **comments** to explain what your code does or to tell your browser what statement(s) to ignore

```
/* This script shows how we can comment lines of code */  
// var yesterday = new Date()           //This statement will be  
ignored  
var today = new Date()                 //This statement will be  
executed
```



JavaScript statement



Multi-line comment



Single-line comment

/VARIABLES

A **variable** allows us to store data temporarily.

EXAMPLE:

```
area = width x height
```

```
area = 5 x 8
```

```
area = 10.0 x 10.0
```

```
area = 1 x "hello"
```

/VARIABLE DECLARATION

VARIABLE KEYWORD

var area;

ASSIGNMENT OPERATOR

var height, width = 0;

VARIABLE VALUE

var customGreeting = "Howdy";

var hidden = false;

VARIABLE NAME

/USING VARIABLES

```
var width, height;  
var area = 0;  
width = 5;  
height = document.getElementById("h").innerHTML;  
area = height * width;
```



/USING STRINGS

```
var viewport;
```

```
var message = "Special Offer";
```

BACKSLASH USED TO
ESCAPE THE
SINGLE QUOTE

```
var user = 'Terry\'s';
```

```
viewport = document.getElementById("view");
```

```
viewport.innerHTML = user + message;
```

+ OPERATOR TO CONCATENATE STRINGS

/STRING FUNCTIONS

```
var message = "Howdy Cowboy";  
var len = message.length;  
var greeting = message.substring(0,5)  
var message = message.replace("Cowboy","Cowgirl")
```

INDEX INCLUDED

⋮

INDEX EXCLUDED

⋮

See more https://www.w3schools.com/js/js_string_methods.asp



/OPERATORS

Operators allow to create a single value from one or more values

ASSIGNMENT

```
color = "#E3E3E3"
```

STRINGS

```
greeting = "Howdy " + "Molly"
```

ARITHMETIC

```
area = 3 * 2;
```

LOGICAL

```
buy = (5>3) && (2<4);
```

COMPARISON

```
limit = quantity >= 10;
```